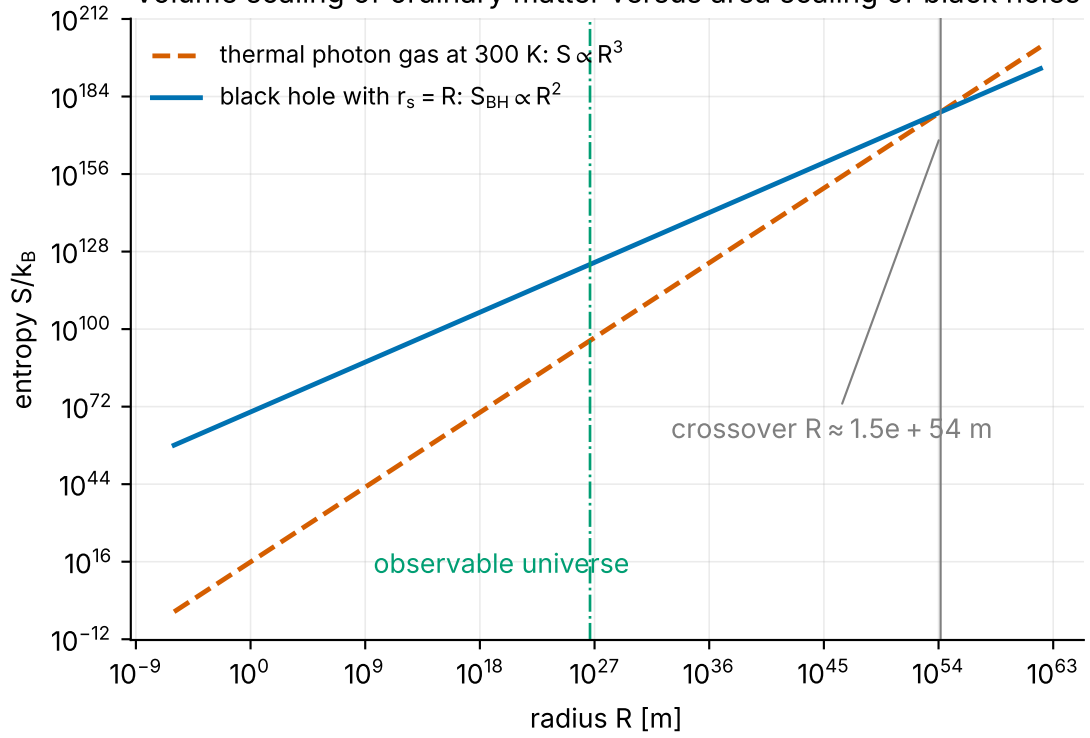
Volume scaling of ordinary matter versus area scaling of black holes
thermal photon gas at 300 K: S ∝ R³
black hole with r_s = R: S_BH ∝ R²
crossover R ≈ 1.5e + 54 m
observable universe
entropy S/k_B
radius R [m]
10²¹²
10¹⁸⁴
10¹⁵⁶
10¹²⁸
10¹⁰⁰
10⁷²
10⁴⁴
10¹⁶
10⁻¹²
10⁻⁹
10⁰
10⁹
10¹⁸
10²⁷
10³⁶
10⁴⁵
10⁵⁴
10⁶³